

Solutions to Second Problem Sheet

on Linear Algebra

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Topic: Algebras

“Algebra is the offer made by the devil to the mathematician. The devil says: I will give you this powerful machine, it will answer any question you like. All you need to do is give me your soul.”
-Michael Atiyah

Problem 1

Show that the cross product $\times : \mathbb{R}^3 \times \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by

$$\mathbf{x} \times \mathbf{y} = \epsilon_{ijk} x_i y_j \mathbf{e}_k, \quad \mathbf{x} = x_i \mathbf{e}_i, \quad \mathbf{y} = y_j \mathbf{e}_j,$$

with $\{\mathbf{e}_k\}_{k=1}^3$ is the canonical basis for $\mathbb{R}^3$, turns it into a non-associative, non-commutative algebra.

Solution 1

We show that the bilinear product $\mathbf{x} \times \mathbf{y} = \epsilon_{ijk} x_i y_j \mathbf{e}_k$ defines on $\mathbb{R}^3$ an algebra which is non-commutative and non-associative. We first investigate commutativity, using the antisymmetry of the Levi-Civita symbol we obtain

$$\mathbf{y} \times \mathbf{x} = \epsilon_{ijk} y_i x_j \mathbf{e}_k = \epsilon_{jik} x_i y_j \mathbf{e}_k = -\epsilon_{ijk} x_i y_j \mathbf{e}_k.$$

Hence,

$$\mathbf{y} \times \mathbf{x} = -(\mathbf{x} \times \mathbf{y}),$$

therefore,

$$\mathbf{x} \times \mathbf{y} \neq \mathbf{y} \times \mathbf{x}$$

in general, so the algebra is non-commutative.

To show non-associativity, consider the canonical basis vectors $\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3$, and using the standard relations

$$\mathbf{e}_1 \times \mathbf{e}_2 = \mathbf{e}_3, \quad \mathbf{e}_2 \times \mathbf{e}_3 = \mathbf{e}_1, \quad \mathbf{e}_3 \times \mathbf{e}_1 = \mathbf{e}_2,$$

we compute

$$(\mathbf{e}_1 \times \mathbf{e}_1) \times \mathbf{e}_2 = 0$$

On the other hand,

$$\mathbf{e}_1 \times (\mathbf{e}_1 \times \mathbf{e}_2) = \mathbf{e}_1 \times \mathbf{e}_3,$$

using antisymmetry,

$$\mathbf{e}_1 \times \mathbf{e}_3 = -\mathbf{e}_3 \times \mathbf{e}_1 = -\mathbf{e}_2$$

thus,

$$(\mathbf{e}_1 \times \mathbf{e}_1) \times \mathbf{e}_2 = 0 \neq -\mathbf{e}_2 = \mathbf{e}_1 \times (\mathbf{e}_1 \times \mathbf{e}_2).$$

Therefore,

$$(\mathbf{x} \times \mathbf{y}) \times \mathbf{z} \neq \mathbf{x} \times (\mathbf{y} \times \mathbf{z})$$

in general, and hence the cross product is non-associative.

Consequently, $(\mathbb{R}^3, \times)$ is an algebra over $\mathbb{R}$ which is non-commutative and non-associative.



Problem 2

The **centre** of an algebra $\mathcal{A}$, denoted by $\mathcal{Z}(\mathcal{A})$ is defined as the set of all elements of $\mathcal{A}$ that commute with all elements of $\mathcal{A}$. Show that the centre of an algebra is a subspace of that algebra. Moreover, show that if the algebra is associative, then its centre is a subalgebra.

Solution 2

Let $\mathcal{A}$ be an algebra over a field $\mathbb{F}$. The centre of $\mathcal{A}$ is defined by

$$\mathcal{Z}(\mathcal{A}) = \{z \in \mathcal{A} \mid za = az, \forall a \in \mathcal{A}\}.$$

Let

$$z_1, z_2 \in \mathcal{Z}(\mathcal{A}), \quad \lambda_1, \lambda_2 \in \mathbb{F}.$$

Then, for every $a \in \mathcal{A}$,

$$(\lambda_1 z_1 + \lambda_2 z_2)a = \lambda_1 z_1 a + \lambda_2 z_2 a = \lambda_1 a z_1 + \lambda_2 a z_2 = a(\lambda_1 z_1 + \lambda_2 z_2).$$

Hence,

$$\lambda_1 z_1 + \lambda_2 z_2 \in \mathcal{Z}(\mathcal{A}),$$

and therefore $\mathcal{Z}(\mathcal{A})$ is a vector subspace of $\mathcal{A}$. Now suppose that $\mathcal{A}$ is associative, let

$$z_1, z_2 \in \mathcal{Z}(\mathcal{A}), \quad a \in \mathcal{A}.$$

Then

$$(z_1 z_2)a = z_1(z_2 a) = z_1(a z_2) = (z_1 a)z_2 = (a z_1)z_2 = a(z_1 z_2)$$

thus

$$z_1 z_2 \in \mathcal{Z}(\mathcal{A}),$$

so $\mathcal{Z}(\mathcal{A})$ is closed under multiplication.

Consequently, if $\mathcal{A}$ is associative, then $\mathcal{Z}(\mathcal{A})$ is a subalgebra of $\mathcal{A}$.



Problem 3

An algebra is said to be **central** if its centre is $\mathcal{Z}(\mathcal{A}) = \text{span}\{\mathbf{1}\}$. Show that the algebra of quaternions is central.

Solution 3

Consider the quaternion algebra

$$\mathbb{H} = \text{span}\{e_0, e_1, e_2, e_3\},$$

with

$$e_1^2 = e_2^2 = e_3^2 = -e_0,$$

and

$$e_1 e_2 = e_3, \quad e_2 e_3 = e_1, \quad e_3 e_1 = e_2,$$

together with

$$e_2 e_1 = -e_3, \quad e_3 e_2 = -e_1, \quad e_1 e_3 = -e_2.$$

We show that

$$\mathcal{Z}(\mathbb{H}) = \mathbb{R}e_0.$$

Let

$$q = q_0 e_0 + q_1 e_1 + q_2 e_2 + q_3 e_3 \in \mathcal{Z}(\mathbb{H}),$$

then it follows

$$q e_1 = e_1 q.$$

Now,

$$q e_1 = (q_0 e_0 + q_1 e_1 + q_2 e_2 + q_3 e_3) e_1 = q_0 e_1 - q_1 e_0 - q_2 e_3 + q_3 e_2,$$

while

$$e_1 q = e_1(q_0 e_0 + q_1 e_1 + q_2 e_2 + q_3 e_3) = q_0 e_1 - q_1 e_0 + q_2 e_3 - q_3 e_2$$

comparing coefficients gives,

$$q_2 = q_3 = 0,$$

hence

$$q = q_0 e_0 + q_1 e_1.$$

Now impose

$$q e_2 = e_2 q$$

so that we obtain

$$q e_2 = (q_0 e_0 + q_1 e_1) e_2 = q_0 e_2 + q_1 e_3,$$

and

$$e_2 q = e_2 (q_0 e_0 + q_1 e_1) = q_0 e_2 - q_1 e_3.$$

Therefore $q_1 = 0$. Thus

$$q = q_0 e_0,$$

which implies

$$Z(\mathbb{H}) = \mathbb{R} e_0.$$

Hence the quaternion algebra is central.



Problem 4

Given an algebra $\mathcal{A}$ and a basis $B = \{\mathbf{e}_i\}_{i=1}^N$ for the underlying linear space, one can write

$$\mathbf{e}_i \mathbf{e}_j = c_{ij}^k \mathbf{e}_k, \quad c_{ij}^k \in \mathbb{C}.$$

The complex numbers c_{ij}^k , i.e., the components of the vector $\mathbf{e}_i \mathbf{e}_j$ in the basis B , are called the **structure constants** of $\mathcal{A}$.

Write down all the structure constants for the algebra of quaternions. Show that this algebra is associative.

Solution 4

Consider the quaternion algebra

$$\mathbb{H} = \text{span}_{\mathbb{R}}\{e_0, e_1, e_2, e_3\},$$

where e_0 is the identity element and

$$e_i^2 = -e_0, \quad e_i e_j = -e_j e_i, \quad i \neq j,$$

with

$$e_1 e_2 = e_3, \quad e_2 e_3 = e_1, \quad e_3 e_1 = e_2.$$

The structure constants $C_{\mu\nu}^{\rho}$ are defined by

$$e_{\mu} e_{\nu} = C_{\mu\nu}^{\rho} e_{\rho}.$$

From the defining relations,

$$\begin{aligned} C_{00}^0 &= 1, & C_{0i}^j &= C_{i0}^j = \delta_i^j, \\ C_{ij}^0 &= -\delta_{ij}, & C_{ij}^k &= \epsilon_{ij}^k, \end{aligned}$$

where ϵ_{ij}^k is the Levi-Civita symbol. Therefore,

$$e_i e_j = -\delta_{ij} e_0 + \epsilon_{ij}^k e_k.$$

Since e_0 is the identity, it suffices to prove associativity for

$$(e_i e_j) e_k, \quad e_i (e_j e_k).$$

Using

$$e_i e_j = -\delta_{ij} e_0 + \epsilon_{ij}^m e_m,$$

we obtain

$$(e_i e_j) e_k = -\delta_{ij} e_k + \epsilon_{ij}^m e_m e_k = -\delta_{ij} e_k + \epsilon_{ij}^m (-\delta_{mk} e_0 + \epsilon_{mk}^n e_n),$$

hence

$$(e_i e_j) e_k = -\epsilon_{ijk} e_0 - \delta_{ij} e_k + \epsilon_{ij}^m \epsilon_{mk}^n e_n.$$

Similarly,

$$e_i (e_j e_k) = -\epsilon_{jki} e_0 - \delta_{jk} e_i + \epsilon_{jk}^m \epsilon_{im}^n e_n.$$

Using

$$\epsilon_{ij}^m \epsilon_{mk}^n = \delta_i^n \delta_{jk} - \delta_{ik} \delta_j^n,$$

and

$$\epsilon_{jk}^m \epsilon_{im}^n = \delta_j^n \delta_{ik} - \delta_{ij} \delta_k^n,$$

we obtain

$$(e_i e_j) e_k = -\epsilon_{ijk} e_0 + \delta_{jk} e_i - \delta_{ik} e_j - \delta_{ij} e_k,$$

and

$$e_i (e_j e_k) = -\epsilon_{jki} e_0 + \delta_{ik} e_j - \delta_{ij} e_k - \delta_{jk} e_i.$$

Since

$$\epsilon_{ijk} = \epsilon_{jki},$$

both expressions coincide, and therefore

$$(e_i e_j) e_k = e_i (e_j e_k).$$

Hence the quaternion algebra is associative. ☆

Problem 5

Show that a quaternion is pure if and only if its square is a non-positive real number.

Solution 5

Suppose $q \in \mathbb{H}$ is pure. Then $q = \mathbf{x}$, with vanishing real part. Using

$$qp = x_0 y_0 - \mathbf{x} \cdot \mathbf{y} + x_0 \mathbf{y} + y_0 \mathbf{x} + \mathbf{x} \times \mathbf{y},$$

and taking $p = q$, we obtain

$$q^2 = -\mathbf{x} \cdot \mathbf{x}$$

Hence

$$q^2 \in \mathbb{R}, \quad q^2 \leq 0.$$

Conversely, suppose q is a non-positive real number and write

$$q = x_0 + \mathbf{x},$$

where $x_0 \in \mathbb{R}$ and $\mathbf{x}$ is pure. Then

$$q^2 = x_0^2 - \mathbf{x} \cdot \mathbf{x} + 2x_0 \mathbf{x}.$$

Since q^2 is real, its pure part must vanish

$$2x_0 \mathbf{x} = 0.$$

Therefore,

$$x_0 = 0 \quad \text{or} \quad \mathbf{x} = 0,$$

if $\mathbf{x} = 0$, then $q = x_0 \in \mathbb{R}$, and thus

$$q^2 = x_0^2 \geq 0,$$

but since $q^2 \leq 0$, we must have $x_0 = 0$ and hence $q = \mathbf{x}$, that is, q is pure.

Therefore, a quaternion is pure if and only if its square is a non-positive real number. ☆

Problem 6

Let $p, q \in \mathbb{H}$. Show that

- i) $(pq)^* = q^*p^*$;
- ii) $q \in \mathbb{R}$ if and only if $q^* = q$, and $q \in \mathbb{R}^3$ if and only if $q^* = -q$;
- iii) $qq^* = q^*q$ is a non-negative real number.

Solution 6

1. Let $q = x_0 + x, p = y_0 + y$, with $x_0, y_0 \in \mathbb{R}$ and x, y pure. Using

$$pq = x_0y_0 - x \cdot y + x_0y + y_0x + x \times y,$$

we obtain

$$(pq)^* = x_0y_0 - x \cdot y - x_0y - y_0x - x \times y.$$

On the other hand, $q^* = x_0 - x, p^* = y_0 - y$, and therefore

$$q^*p^* = x_0y_0 - x \cdot y - x_0y - y_0x - x \times y = (pq)^*.$$

2. Suppose first that $q = \alpha \in \mathbb{R}$. Then $q^* = \alpha^* = q$. Conversely, let $q = x_0 + x$ and assume that $q^* = q$. Since $q^* = x_0 - x$, we have $x_0 + x = x_0 - x$, hence $x = 0$, and therefore

$$q = x_0 \in \mathbb{R}.$$

3. Let $q = x_0 + x$ and $q^* = x_0 - x$. Then

$$qq^* = (x_0 + x)(x_0 - x) = x_0^2 - x \cdot (-x) - x_0x + x_0x + x \times (-x).$$

This reduces to

$$qq^* = x_0^2 + |x|^2.$$

Similarly,

$$q^*q = x_0^2 + |x|^2,$$

therefore

$$qq^* = q^*q = x_0^2 + |x|^2 \in \mathbb{R}, \quad qq^* \geq 0.$$

